

Match-3 Take-Home Assignment (v3)

Welcome to the Match-3 game assignment! You will implement a falling bricks game. In this game, you will control “bricks” made up of three blocks, each marked with a special symbol. Bricks can be horizontal or vertical, and you can move or drop them using simple commands. The goal is to place all bricks and clear as many matches as possible.

Initialization

- On startup, prompt the user:
 - > Please enter field size (width and height) and up to 5 bricks set:
 - >
 - > Example: `5 8 H^^\* V\*@^`
- The first two numbers are the field width and height.
- Each brick is defined by an orientation (`H` for horizontal, `V` for vertical) followed by three symbols (e.g., `H^^\*`).
- Allowed symbols: `~`, `^`, `\*`, `@`.
- Up to 5 bricks can be provided.

Game Loop

1. The first brick appears at the top of the field:
 - Horizontal: centered in the top row.
 - Vertical: centered in the top three rows.
 - If the starting position is blocked, the game ends immediately.
2. Each frame:
 - Display the current field and active brick.
 - Prompt the user:
 - > Enter up to 2 commands to process before moving to the next frame (valid commands are L,R,D):
 - Only the first two valid commands are processed per frame.
 - Commands:
 - `L`: Move the brick left by one column (if possible)
 - `R`: Move the brick right by one column (if possible)
 - `D`: Drop the brick as far down as possible, stopping above any existing brick or at the bottom
 - After commands, the brick always drops by one row automatically.
 - If a command would move the brick out of bounds or into another brick, it is ignored.
 - If the brick cannot move further, it becomes stationary.
 - When a brick becomes stationary:
 - Check for any horizontal or vertical line of 3 or more matching symbols and remove them.
 - Removed cells become empty; gravity is not applied.
 - All matches can be removed at once.
 - The next brick appears at the top. If it cannot be placed, the game ends.

3. Repeat until all bricks are used or no more can be placed.

Game End

- After the game ends, prompt:
 - > Enter S to start over or Q to quit:
- > - `S`: Restart the game from initialization.
- > - `Q`: Exit the game.
- Display a thank-you message upon exiting:
 - > Thank you for playing Match-3!

Example Scenario

...

Welcome to Match-3 game!

Please enter field size (width and height) and up to 5 bricks set:

5 8 H^^* V*@^

Frame 1

```
| . ^ ^ * . |
| . . . . . |
| . . . . . |
| . . . . . |
| . . . . . |
| . . . . . |
| . . . . . |
| . . . . . |
```

Enter up to 2 commands to process before moving to the next frame (valid commands are L,R,D):

LL

Frame 2

```
| . . . . . |
| ^ ^ * . . |
| . . . . . |
| . . . . . |
| . . . . . |
| . . . . . |
| . . . . . |
| . . . . . |
```

Enter up to 2 commands to process before moving to the next frame (valid commands are L,R,D):

R

Frame 3

```
| . . . . . |
```


.	.	.	.	.
.	^	^	*	.
.	.	.	.	.
.	.	.	.	.
.	.	.	.	.
.	.	.	.	.

Enter up to 2 commands to process before moving to the next frame (valid commands are L,R,D):
 DR

Frame 4

.	.	*	.	.
.	.	@	.	.
.	.	^	.	.
.	.	.	.	.
.	.	.	.	.
.	.	.	.	.
.	.	^	^	*

Enter up to 2 commands to process before moving to the next frame (valid commands are L,R,D):
 LLR

Frame 5

.	.	.	.	.
*	.	.	.	.
@	.	.	.	.
^	.	.	.	.
.	.	.	.	.
.	.	.	.	.
.	.	^	^	*

Enter up to 2 commands to process before moving to the next frame (valid commands are L,R,D):

Frame 6

.	.	.	.	.
.	.	.	.	.
*	.	.	.	.
@	.	.	.	.
^	.	.	.	.
.	.	.	.	.
.	.	.	.	.
.	.	^	^	*

Enter up to 2 commands to process before moving to the next frame (valid commands are L,R,D):
 R

Frame 7

.	.	.	.	.
---	---	---	---	---


```

| . . . . . |
| . . . . . |
| . * . . . |
| . @ . . . |
| . ^ . . . |
| . . . . . |
| . . ^ ^ * |

```

Enter up to 2 commands to process before moving to the next frame (valid commands are L,R,D):
DR

Frame 8

```

| . . . . . |
| . . . . . |
| . . . . . |
| . . . . . |
| . * . . . |
| . @ . . . |
| . . . . * |

```

Game Over.
Enter S to start over or Q to quit:
Q

Thank you for playing Match-3!
` ``

Example Scenario Explanation

Frame	Remarks
1	The first horizontal brick ^^* is rendered on the top row. User enters LL command.
2	Only the first L command is processed as the second L command would bring the brick outside the field. The brick dropped 1 row, so it is at 2nd row from the top now. User then enters R command.
3	The brick is moved to the right, and dropped 1 row. It is at the 3rd row from the top now. User then enters DR command.
4	The brick is dropped to the bottom of the field, moved to the right and dropped 1 row, which caused the brick to stop moving. As it cannot drop to any lower row, the second brick *@^ is rendered starting at the top row. User then enters LLR command.
5	The brick can only move to the left once, as the next L command would bring the brick outside the field, and the R command is ignored as program can only process 2 commands in one frame. User then enters blank command.
6	The brick dropped 1 row without moving left or right. User then enters R command.
7	The brick is moved to the right, and dropped 1 row. User then enters DR command.
8	The brick is dropped to the bottom row, but as there is an adjacent brick,

the brick could not move to the right. The brick cannot move down anymore, so program checks for matching symbols. As there are 3 ^ symbols horizontally, program removes them from the field. As there are no more bricks to process, a game over message is shown. |

finalight@gmail.com 2026-02-02CONFIDENTIAL DO NOT SHARE